
Agentic Geometry Problem Solving via Human-like Parallel Bidirectional Reasoning

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Abstract

Formal Geometric Problem Solving requires every step of reasoning within a strict logical framework, and has been a core challenge in artificial intelligence. This paper presents a unified neuro-symbolic reasoning framework, named **Reflect Agent Verified Solve (RAVS)**, which deeply integrates large language models, agent architectures, and formal symbolic solvers. The core innovations of RAVS lie in two parts. First, we propose a role-separated neuro-symbolic collaboration mechanism: the LLM serves as a planner responsible for high-level semantic understanding and path reflection, while the symbolic solver acts as an executor responsible for formal verification and rigorous theorem application. The neural reasoning capability of the LLM and the logical completeness of the symbolic system complement each other, fundamentally eliminating the risk of hallucinations. Second, we construct the first bidirectional symbolic reasoning engine that fully unifies forward and backward solving. For each theorem, we define two reversible operations: forward application and backward decomposition. We further provide a theoretical analysis that bidirectional reasoning achieves an exponential complexity advantage over unidirectional reasoning. Together, these two mechanisms form an iterative closed loop between neural planning and symbolic execution. On the FormalGeo7K benchmark, RAVS achieves a 96.06% solving accuracy, substantially outperforming existing state-of-the-art methods, without requiring any additional problem-specific annotated data. Code and data are available at <https://github.com/BitSecret/RAVS>.

1 Introduction

Geometric Problem Solving has long been a fundamental challenge [26, 20] in artificial intelligence and automated reasoning. **Formal Geometric Problem Solving** requires every step of reasoning to be conducted within a strict logical framework and rigorously verified. This makes it considerably more difficult than solving geometry problems using the natural language. Specifically, the task faces two key challenges. First, geometry problems involve both image understanding and text parsing. The solving process interweaves relational reasoning, algebraic computation, and logical operations, with reasoning strategies encompassing deductive reasoning and goal decomposition. Formalizing this process in its entirety is inherently difficult. Second, the theorem search space grows exponentially with the number of reasoning steps, and the reasoning process can be represented as a large directed acyclic graph with multiple potential solution paths, further compounding the search complexity.

The research community has pursued this problem for decades, progressing through several stages: search-based methods [19, 31], computation-based methods [45, 3], geometric expert systems [11, 64], and approaches based on machine learning [39, 38]. Before the rise of deep learning, geometric reasoning primarily relied on manually designed rules and search heuristics, lacking the ability to

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acquire knowledge automatically. In recent years, two mainstream technical paradigms have emerged for geometric problem solving. The first paradigm [59, 61, 17, 13] is natural language reasoning driven by general-purpose large language models (LLMs), which directly generates natural language solution steps. The second paradigm [41, 29, 62, 21, 58] combines specialized theorem prediction neural networks with formal symbolic solvers: the neural network predicts the theorem required for the solution, while the symbolic solver performs precise reasoning and computation. These two components interact to achieve efficient heuristic search.

However, the first paradigm suffers from a fundamental problem: natural language proofs generated directly by LLMs cannot guarantee correctness. Even with enhanced frameworks such as CoT [43] or ReAct [54], the generated steps may violate the premise of geometric theorems. The absence of a formal verification mechanism makes such approaches difficult to deploy in scenarios requiring high reliability. To fundamentally address this issue, this paper proposes **Reflect Agent Verified Solve (RAVS)**. The design of RAVS follows a core principle: let the LLM do what it excels at (high-level planning and abstract reasoning), and let the symbolic system do what it excels at (rigorous verification and precise computation). During the solving process, the LLM analyzes the current state, reflects on previously explored paths, and formulates the next plan, while the actual theorem application and algebraic computation are performed by a formal symbolic solver. Every operation in the system must pass strict verification by the solver and receive feedback, thereby eliminating hallucinations at the architectural level. In particular, RAVS introduces a memory update mechanism: when the context reaches a certain size, the LLM automatically summarizes the progress, identifies failed exploration paths, and dynamically updates its solving strategy accordingly. This closed loop of "reasoning, execution, reflection, and memory" enables the system to maintain an efficient reasoning direction on complex problems.

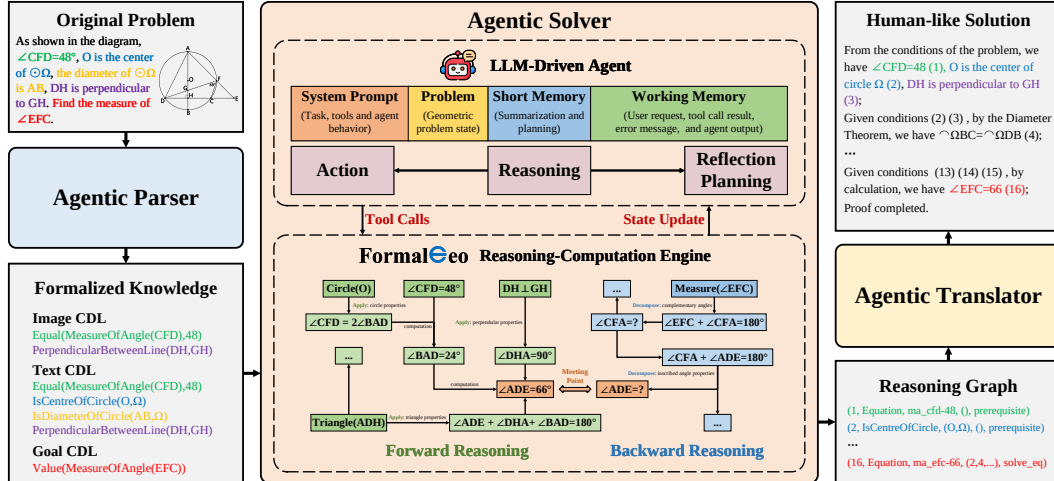


Figure 1: Overall Architecture of **Reflect Agent Verified Solve (RAVS)**. Geometric problems are parsed into formal language by an Agentic Parser, then input into an Agentic Solver for solving. After the problem is solved, the reasoning graph is transformed into a human-readable solution by an Agentic Translator. The Agentic Solver is the core method introduced in this paper. It consists of an LLM-driven agent (high-level planning and abstract reasoning) and a formal bidirectional solver based on FormalGeo (rigorous verification and precise computation).

On the other hand, while the second paradigm guarantees the correctness of reasoning, its efficiency is limited by inherent shortcomings of existing formal systems [11, 27, 5, 63]. Humans employ two modes of thinking when solving geometry problems: forward reasoning (deriving conclusions from given conditions) and backward reasoning (decomposing the goal into subgoals). Backward reasoning is often more efficient as the goal provides rich heuristic information. However, existing systems all adopt forward reasoning and have never considered the implementation of reversible operations at the formal theorem level. Given a reasoning depth d and a theorem library of size n , the worst-case complexity of search is $O(n^d)$. This exponential explosion renders such systems inadequate for problems with long reasoning chains. The bidirectional reasoning engine proposed in this paper fundamentally changes this situation. During reasoning, the system simultaneously

searches in both forward and backward direction, and terminates when the two search frontiers meet at some intermediate state. Theoretical analysis also shows that bidirectional reasoning is superior to any unidirectional approach. Furthermore, the entire solving process is organized into a clear dual-graph structure: the forward constructs a hypergraph of facts and theorems, while the backward constructs a composite graph of goals, subgoals, and theorems. This structure not only makes the reasoning process transparent but also can be naturally converted into human-readable solution steps.

In summary, this paper proposes the third technical paradigm for geometric problem solving in the deep learning era: a deep integration of large language models, agent architectures, and formal symbolic solvers. The overall framework is illustrated in Figure 1. In this framework, the LLM does not directly participate in symbolic-level reasoning; instead, it acts as a high-level planner that provides strategic guidance to the formal symbolic solver. The symbolic solver is responsible for executing rigorous formal reasoning and computation, ensuring complete correctness of the entire solving process at the architectural level. Compared with the two existing paradigms, our framework can simultaneously leverage the general knowledge and strong planning capabilities of LLMs while achieving fully reliable reasoning through the correctness verification provided by the symbolic system. Extensive experiments validate the effectiveness of the proposed method. RAVS achieves a 95.80% accuracy rate on the FormalGeo7K [65] benchmark, significantly surpassing existing state-of-the-art methods. We also provide three examples to demonstrate the advantages of RAVS, as shown in Figure 2.

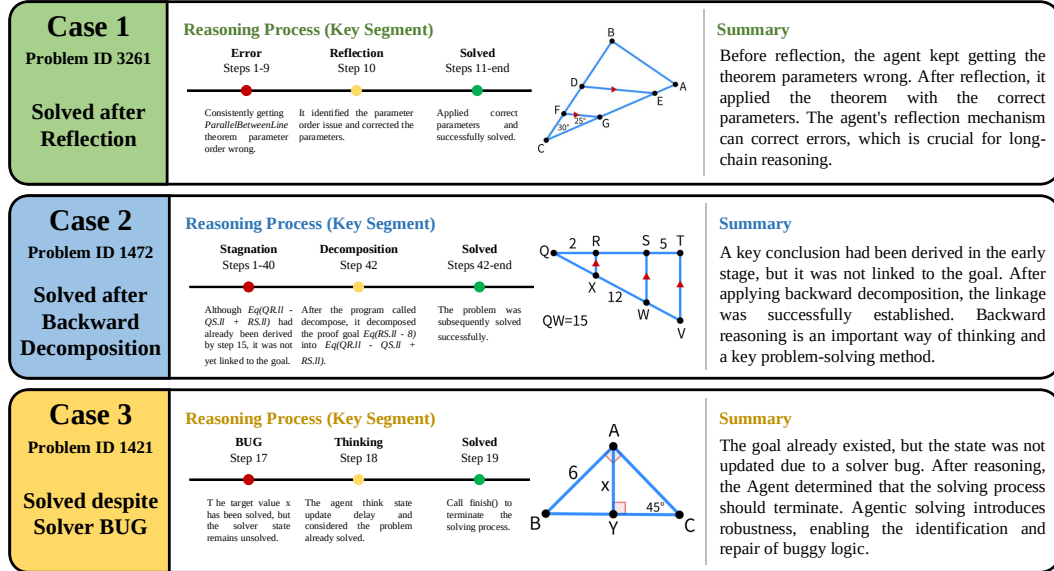


Figure 2: Three examples illustrating the advantages of RAVS. **Case 1** illustrates that the agent’s reflection mechanism helps correct errors in long-chain reasoning. **Case 2** illustrates that backward solving is an important solving method. **Case 3** illustrates that the RAVS architecture can correct bugs in the solver and offers greater flexibility.

The core innovations and contributions of this paper are summarized as follows:

1. We propose the RAVS framework, a new technical paradigm that deeply integrates large language models, agent architectures, and formal symbolic solvers. By adopting a decoupled design where the LLM acts as a high-level planner and the symbolic solver performs rigorous verification, this approach achieves verifiable, hallucination-free formal geometric problem solving.
2. We introduce a bidirectional formal symbolic reasoning mechanism, which defines two reversible operations: forward reasoning (deriving conclusions from given conditions) and backward reasoning (decomposing the goal into subgoals). We theoretically analyze that bidirectional reasoning offers an exponential complexity advantage over unidirectional reasoning.
3. We conduct experiments to validate the effectiveness of the proposed method. On the FormalGeo7K [65] benchmark, our method achieves a 95.80% accuracy rate, significantly surpassing existing state-of-the-art methods Pri-TPG [67] 89.29%, without relying on any additional labeled data for specific problems.

2 Related work

Geometry is a discipline that studies numbers and shapes. Based on the different fundamental elements they process, we classify geometric problem solving methods into three categories: algebraic methods, hybrid methods, and geometric methods.

Algebraic methods [46, 3, 52] transform geometric problems into algebraic form, enabling solution via algebraic techniques. Wu’s method [45] was a foundational contribution, followed by Buchberger’s Gröbner bases [3], numerical parallel methods [51], polynomial triangulation elimination [18], cylindrical algebraic decomposition [12], dimensionality reduction [52]. While these methods leverage computational power, their solving processes are not easily interpretable to humans.

Hybrid methods [60, 9] bridge algebra and geometry to produce more interpretable proofs than purely algebraic approaches, though their problem coverage is limited. Point elimination method [60] uses constructive descriptions based on geometric invariants, generating concise readable proofs for many non-trivial problems. This line of work advanced rapidly [7, 9, 23], producing tools such as Geometry Explorer [44], Geometry Expert [10], and Java Geometry Expert [55], with extensions to solid [8] and non-Euclidean [53] geometry. However, the method demands substantial effort in discovering new invariants and shape constructions, limiting the range of solvable problems.

Geometric methods have been the most extensively studied, progressing through three notable stages: early symbolic reasoning [19, 31], geometric expert systems [11, 64], and machine learning-based approaches [39, 1, 38, 56]. More recently, **deep learning-based methods** have advanced the field along two routes. The first uses general-purpose LLMs for natural language reasoning [59, 16, 28, 57, 61, 25, 17, 66, 13]. This avoids building a formal system but cannot verify output correctness. The second combines specialized theorem prediction networks with formal symbolic solvers [41, 29, 62, 21, 58], using formal language (or explainable operations [49, 47, 48, 42, 32], or Python code [40]) for reasoning. This builds more reliable systems but the cost of human annotation or synthetic formal data is high.

3 Preliminaries

3.1 Problem Definition

Definition 1 (Geometric Problem). A geometric problem is defined as a tuple $Q = (F, G, E)$. Here, $F = \{f_1, f_2, \dots, f_m\}$ is a set of known facts, $G = \{g_1, g_2, \dots, g_n\}$ is a set of goals to be proved, where the first goal g_1 is the original solving target and the remaining goals are those derived through decomposition. Each goal is a geometry predicate logic expression tree composed of subgoals s_i and logical connectives (AND, OR, and NOT). Each fact and each subgoal is represented by a geometric predicate. $E = \{E^{(f)}, E^{(g)}, E^{(s)}\}$ denotes the set of edges connecting facts, goals, and subgoals.

Definition 2 (Theorem). A theorem is defined as a tuple $\tau = (P, C)$, where P is the premise, expressed as a geometric predicate logic expression composed of geometric predicates and logical connectives, and C is the conclusion, consisting of a single geometric predicate. The theorem library $T = \{\tau_1, \tau_2, \dots, \tau_k\}$ is the set of all available theorems.

3.2 Task Definition

Let $Q_i = (F_i, G_i, E_i)$ denote the state of a geometric problem at time step i . The task of geometric solving is to find an ordered sequence of theorems $S = [\tau_1, \tau_2, \dots, \tau_t]$ from the theorem library T , along with a corresponding sequence of application modes $U = [u_1, u_2, \dots, u_t]$, such that applying these theorems in order proves the initial goal g_1 , i.e., $g_1 \in F_t$. Each theorem $\tau_j = (P_j, C_j)$ can be applied in two modes:

Definition 3 (Forward Solving). If the geometric predicate logic expression P_j evaluates to true given the current fact set F_i , then C_j is added to F_i : $F_{i+1} \leftarrow F_i \cup \{C_j\}$. The theorem τ_j is also added to the edge set: $E_{i+1} \leftarrow E_i \cup \{\tau_j\}$.

Definition 4 (Backward Solving). If C_j is a subgoal of some current goal $g \in G_i$, then P_j is added to the goal set G_i : $G_{i+1} \leftarrow G_i \cup \{P_j\}$. The theorem τ_j is also added to the edge set: $E_{i+1} \leftarrow E_i \cup \{\tau_j\}$.

3.3 Interaction Process

The previous subsections describe how a geometric problem is represented and solved within the solver. We now formally define the interaction process between the solver and the agent. We model this interactive solving process as a quadruple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \pi)$.

State Space $\mathcal{S}$. The state at step t is defined as $s_t = (Q_t, M_t)$, where Q_t is the current problem state and M_t is the agent’s memory.

Action Space $\mathcal{A}$. An action $a_t \in \mathcal{A}$ consists of selecting a theorem $\tau \in T$ and specifying an application mode $u \in \{\text{forward}, \text{backward}\}$. An action is valid only if the corresponding operation is executable in the current state.

State Transition Function $\mathcal{P}$. The transition $s_{t+1} = \mathcal{P}(s_t, a_t)$ is deterministic. After executing action a_t , the solver updates the state according to the predefined semantics of the theorem, producing a unique successor state.

Policy π . The policy $\pi(a_t \mid s_t; w, M_t)$ is parameterized by the LLM weights w and the memory module M_t , where M_t stores the history of reasoning and reflection.

The agent selects a theorem from the action space $\mathcal{A}$ according to the policy π . The solver applies the theorem, updates the problem state via the state transition function $\mathcal{P}$, and returns feedback to the agent. The agent incorporates this feedback into its memory, updates the state space $\mathcal{S}$, and proceeds to the next iteration until the problem is solved.

4 Method

4.1 Bidirectional Reasoning

We implement a formal bidirectional symbolic solver for geometric problems using Python, built upon the FormalGeo formal system. The solver uniformly represents the relational reasoning, algebraic computation, and logical operations involved in geometric problem solving as geometric predicate logic operations, and unifies forward and backward solving within a single formal framework.

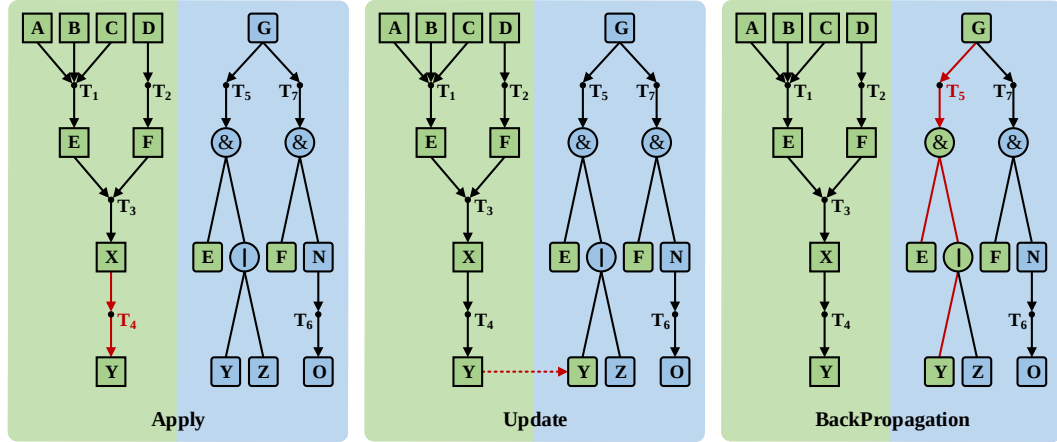


Figure 3: Reasoning graph of the bidirectional solving process. The green graph represents the forward reasoning graph, while the blue graph represents the backward reasoning graph. **Step 1: Apply.** The forward solver selects an applicable theorem from the current fact set, derives new facts, and updates the fact set. **Step 2: Update.** Whenever a new fact is produced, the system immediately queries a hash table that records all pending subgoals in the backward process. If a matching subgoal exists, its state is updated to solved. Since the hash table lookup has a time complexity of $O(1)$, the overhead of meeting detection is negligible. **Step 3: Propagation** After a subgoal state is updated, the system propagates the change upward according to logical operation rules. If all subgoals of a goal are marked as solved, the goal’s state is updated to solved, and the theorem used to decompose the goal is fed back to the forward process, triggering a new round of interaction.

Humans typically employ two modes of thinking when solving geometric problems: forward reasoning (deriving conclusions from given conditions) and backward reasoning (decomposing the goal into subgoals). In certain scenarios, backward reasoning is more efficient because the goal itself provides rich heuristic information. Although the idea of backward reasoning emerged early [31] in the field of geometric reasoning, it has received far less attention than forward reasoning. Recent systems [64] that implement backward reasoning still treat forward and backward reasoning as independent modules rather than integrating them into a unified framework. Compared with forward reasoning, backward reasoning is more difficult to implement due to its complex state updates. Building a deeply integrated system that unifies both directions poses an even greater challenge. The core difficulties lie in two aspects: how to describe both reasoning modes in a unified manner, and how to accurately detect when forward and backward meet.

This paper presents the first fully unified forward and backward reasoning system for geometric problem solving. The mechanisms for meeting detection, interactive feedback, and state updates in backward solving are illustrated in Figure 3. The iterative interaction process of bidirectional reasoning is fully automatic and requires no human intervention. Through this unified bidirectional reasoning framework, the system simultaneously leverages the completeness of forward reasoning and the heuristics of backward reasoning, significantly improving search efficiency while maintaining correctness.

We theoretically show that bidirectional reasoning outperforms any unidirectional approach. Viewed as heuristic search with an LLM agent providing guidance, let D denote problem difficulty, i.e., the length of the theorem sequence needed for solving. The search tree grows exponentially with D , so larger D means harder problems.

For a problem of difficulty D , consider forward, backward, and bidirectional search. In forward search, each node expands to M children, with time a per expansion. The time to reach depth d is

$$T_{forward} = a \sum_{i=1}^d M^i = a(M^1 + M^2 + \dots + M^d)$$

In backward search, each node expands to N children, with time b per expansion. From depth D toward depth 0, the time to cover the remaining distance $D - d$ is

$$T_{backward} = b \sum_{j=1}^{D-d} N^j = b(N^1 + N^2 + \dots + N^{D-d})$$

Here M , N , a , and b depend on the theorem library: more theorems yield larger values. D reflects problem complexity, while M , N , a , and b reflect formal system complexity.

Bidirectional search outperforms unidirectional search under the following assumptions: expanding the last forward (or backward) layer takes more time than expanding the first backward (or forward) layer:

$$a \cdot M^D > b \cdot N \quad \text{and} \quad b \cdot N^D > a \cdot M$$

Then there exists a depth d where forward search is more efficient before d , and backward search after d . This switching point satisfies

$$M^d \leq N^{D-d} \quad \text{and} \quad M^{d+1} \geq N^{D-d-1}$$

Furthermore, traditional backward reasoning operates at the granularity of predicates (e.g., decomposing the goal `RightTriangle(A,B,C)`), while forward reasoning operates at the granularity of theorems (e.g., applying the theorem `pythagorean_inverse` to determine whether a `Triangle(A,B,C)` is right-angled). This mismatch in granularity makes it difficult to unify forward and backward reasoning. To address this, we explicitly require that the conclusion of each theorem consists of a single predicate. Then each theorem is associated with a specific conclusion (goal). Consequently, we can describe forward and backward reasoning in a unified manner: both apply a theorem to either expand facts or decompose goals.

Algorithm 1 RAVS Solving Loop

Require: Geometric Problem $Q = (F, G, E)$, Theorem Library T , Maximum Epochs $E_{\max}$
Ensure: Solving Result $status \in \{\text{solved}, \text{failed}\}$, Solving Trajectory M_t

- 1: Initialize symbolic solver SOLVER with theorem library T and problem Q
- 2: Initialize LLM Agent AGENT with system prompt and solver state SOLVER.state()
- 3: $e \leftarrow 0$
- 4: **while** $e < E_{\max}$ **do**
- 5: $e \leftarrow e + 1$
- 6: $response \leftarrow \text{AGENT.reasoning}()$ ▷ Step 1: Reasoning
- 7: $(tool, args) \leftarrow \text{ParseTool}(response)$
- 8: **if** $tool \neq \text{summarize}$ **then**
- 9: $result \leftarrow \text{ExecuteTool}(\text{SOLVER}, tool, args)$ ▷ Step 2: Tool Execution
- 10: **else**
- 11: $\text{AGENT.summarize}(args, \text{SOLVER.state}())$ ▷ Step 3: Reflection and Planning
- 12: **continue**
- 13: **end if**
- 14: $\text{AGENT}.M_t \leftarrow \text{AGENT}.M_t \cup [response, result]$ ▷ Step 4: Memory Update
- 15: **if** $\text{SOLVER.status}() = \text{solved}$ **then** ▷ Step 5: Termination Check
- 16: **break**
- 17: **end if**
- 18: **end while**
- 19: **return** $\text{SOLVER.status}(), \text{AGENT}.M_t$ ▷ Step 6: Result Determination

4.2 Agentic Loop

Existing LLM-based geometric solving methods typically require the LLM to directly generate complete proof steps, including theorem selection, algebraic computation, and the construction of reasoning chains. However, LLMs inevitably produce logical gaps or premise violations in long-sequence reasoning—a phenomenon known as "hallucination". Even with enhanced frameworks such as CoT [43] or ReAct [54], the lack of formal verification mechanisms makes such approaches difficult to deploy in high-reliability scenarios.

This section details the solving loop of RAVS. The design of RAVS follows a core principle: let the LLM do what it excels at (high-level planning and abstract reasoning), and let the symbolic system do what it excels at (rigorous verification and precise computation). This principle is realized as a four-stage closed loop "reasoning, execution, reflection, and memory", as shown in Algorithm 1:

Stage 1: Reasoning. At each loop iteration, the LLM generates reasoning based on the memory buffer M_t , which contains the system prompt, initial problem state, and full interaction history.

Stage 2: Tool Execution and Reflection (Steps 2-3). The LLM's response is parsed into a tool call. RAVS defines two core operations: forward application (`apply`) to derive new facts, and backward decomposition (`decompose`) to split a goal into subgoals. Auxiliary tools include `find_fact`, `find_goal`, and `check`. Two special control tools are provided: `summarize` triggers reflection when a bottleneck is reached, compressing the explored trajectory into a structured summary to replace redundant history; and `finish` actively terminates the solving loop when the goal is proved or deemed unsolvable.

Stage 3: Memory Update (Step 4). Tool execution results and error messages are appended to M_t , ensuring the LLM maintains complete context. A context length threshold $L_{\max}$ triggers automatic summarization when exceeded.

Stage 4: Termination Decision (Steps 5-6). At each loop end, the system checks the solver's goal status: early termination with success if proved, timeout if $E_{\max}$ is reached, or termination via `finish` with status determined by the solver.

5 Experiment

5.1 Datasets and Evaluation

We evaluate on three formal geometry benchmarks: FormalGeo7K [65], Geometry3K [27], and GeoQA [5]. Geometry3K contains 3,002 diagram-text geometry problems with formal language parsing annotations. GeoQA provides 5,010 geometry problems with annotated programs that describe explicit solving procedures. FormalGeo7K comprises 7,000 geometry problems and has recently received widespread attention from the geometry problem-solving community [62, 30, 6, 35, 67]. It is a large-scale formal geometry dataset with natural language descriptions, high-quality geometric diagrams, GeoGebra-formatted geometric image source files, ground-truth formal language annotations, theorem sequence and theorem directed acyclic graph annotations, designed to support various tasks such as geometry problem parsing and solving. We split the dataset into training, validation, and test sets in a 4:1:1 ratio, and report our method’s performance on the test set. Following FGeo-HyperGNet [62], we stratify test problems into six difficulty levels (L1–L6): L1 ($1 \leq l \leq 2$), L2 ($3 \leq l \leq 4$), L3 ($5 \leq l \leq 6$), L4 ($7 \leq l \leq 8$), L5 ($9 \leq l \leq 10$), and L6 ($l \geq 11$), where l is the number of theorems required to solve the problem. We report accuracy (%) as the proportion of solved problems in the test set. A problem is considered solved if and only if the formal solver state is solved.

5.2 Benchmark Methods

We compare our method against representative baselines. **Symbolic-only solvers** employ search-based approaches to solve problems, using the relationship between the problem’s known conditions as heuristic information, combined with various search strategies (e.g., depth-first search, breadth-first search) to solve problems via heuristic search. **Neural-only solvers** solve geometry problems in an end-to-end fashion using LLMs. These methods directly generate natural language solutions, and the evaluation of solutions is also performed through LLMs. **Neural-symbolic solvers** integrate neural networks with a symbolic reasoning engine to perform theorem-driven search. They can be further divided into training-based and training-free. **Training-based neural-symbolic solvers** fine-tune a pre-trained neural network or train a theorem prediction network from scratch to perform theorem selection, solving problems through the interaction between a theorem predictor and a symbolic solver. **Training-free neural-symbolic solvers** leverage pre-trained LLMs for theorem prediction and apply theorems through symbolic solvers, avoiding any task-specific training.

5.3 Experiment Setup

Our method runs comfortably on a personal laptop (CPU i7-13650HX, RAM 16GB, GPU RTX 4070). We set the maximum number of interaction rounds to 50 to avoid infinite loops, and the maximum memory length to 80k to force reflection and replanning when the model does not reflect autonomously. We use multi-processing to improve CPU utilization and API call efficiency. For 110 problems that were not solved in the first round of interaction, we attempt to solve them again a second time.

We selected three base models: Qwen3.6-Plus [37], MiMo-V2.5-Pro [50], and DeepSeek-V4-Pro [15]. Qwen3.6-Plus consumes 1M tokens to attempt 0.77% of the problems, MiMo-V2.5-Pro consumes 200M tokens to attempt 15.6% of the problems, and DeepSeek-V4-Pro consumes 370M tokens to attempt 83.63% of the problems.

5.4 Experimental results

Table 1 presents the performance various methods on the FormalGeo7K dataset [65]. Our proposed method, RAVS, achieves state-of-the-art performance with an overall accuracy of 95.80%, significantly outperforming every baseline method across all difficulty levels.

Purely symbolic methods are fundamentally limited on complex problems. Purely symbolic solvers guarantee reasoning correctness but achieve only 33%–40% overall accuracy. Forward search outperforms backward search in both overall accuracy and across all difficulty levels, yet both methods drop below 10% on L5 and L6 problems, revealing their inherent weakness in long-chain reasoning without high-level guidance.

Table 1: Comparison results between different methods on FormalGeo7K (*accuracy, %*). Our proposed method, RAVS, achieves state-of-the-art performance, significantly outperforming every baseline method across all difficulty levels.

Methods	Total	L1	L2	L3	L4	L5	L6
Symbolic-Only Methods							
Backward-DFS [64]	33.73	67.99	32.99	9.78	5.90	5.36	0.62
Backward-RS [64]	34.05	68.85	33.51	9.86	5.22	4.76	0.62
Forward-BFS [64]	38.86	61.37	37.89	32.33	17.91	9.23	2.49
Forward-RS [64]	39.71	60.74	39.33	37.22	16.55	6.25	4.05
Neural-Only Methods							
Kimi-K2 [22]	60.75	69.05	70.29	65.77	36.00	33.33	19.15
DeepSeek v3 [14]	60.79	75.72	58.81	63.26	46.29	23.44	33.87
GPT-5 mini [34]	64.79	74.11	63.30	64.66	53.50	53.23	41.46
Qwen3-VL [36]	65.93	74.53	65.43	72.18	50.96	41.94	36.67
Doubao seed 1.8 [4]	69.14	74.11	69.15	71.43	64.33	50.00	51.67
GPT-5.2 [33]	73.14	80.38	73.40	74.81	63.06	59.68	46.67
Claude4.5 Sonnet [2]	75.79	84.55	73.94	76.32	67.52	64.52	48.33
Neural-Symbolic Methods (Training-Based)							
Inter-GPS [27]	60.50	81.07	60.62	60.98	38.86	20.31	11.29
DualGeoSolver [49]	62.11	63.64	68.10	65.94	58.00	60.98	31.71
NGS [5]	62.60	62.88	65.64	71.01	57.00	60.98	36.59
FGeo-DRL [68]	80.85	97.63	92.11	70.89	61.60	39.39	31.37
FGeo-ISRL [24]	85.16	95.12	93.30	82.26	73.24	61.90	35.20
HyperGNet [62]	88.36	96.44	92.49	91.29	78.29	57.81	51.61
Neural-Symbolic Methods (Training-Free)							
Pri-TPG [67]	89.29	99.16	96.28	87.92	77.07	66.13	30.00
Ours (RAVS)	96.06	99.74	98.72	94.59	96.00	91.23	61.70

Table 2: Comparison results between different methods on Geometry3K, GeoQA and FormalGeo7K (*accuracy, %*). Our method achieves the best reported accuracy on every datasets. – indicates results not reported. * indicates evaluation on the subset parsed into the FormalGeo formal language.

Method	Geometry3K	GeoQA	FormalGeo7K
E-GPS [47]	90.40	–	–
DualGeoSolver [49]	–	65.20	–
FGeo-ISRL [24]	–	–	85.16
HyperGNet* [62]	91.99	85.64	88.36
Pri-TPG* [67]	95.16	85.02	89.29
Ours (RAVS)*	98.11	94.64	96.06

Training-based neural-symbolic methods still struggle to generalize to hard problems. State-of-the-art training-based methods like FGeo-ISRL (85.16%) and HyperGNet (88.36%) perform well on simpler problems but show significant accuracy decline on L5–L6, with HyperGNet dropping to 51.61% on L6. Purely LLM-based methods achieve over 80% on L1 but also suffer from performance degradation on hard problems and lack formal verification, making their results unreliable.

RAVS achieves superior and consistent performance across all difficulty levels. RAVS reaches 95.80% overall accuracy, with particularly striking gains on L5 (91.23% vs. 66.13%) and L6 (59.57% vs. 51.67%). Without any training, our reflection-driven neuro-symbolic framework also outperforms the training-free LLM baseline Pri-TPG by 6.51 percentage points, demonstrating that the integration of LLM planning with symbolic verification, mediated through bidirectional reasoning and agentic reflection, delivers robust and verifiable formal reasoning across the entire difficulty spectrum. However, RAVS’s performance on L6-level difficulty is limited.

Table 2 report comparison results between different methods on Geometry3K, GeoQA and FormalGeo7K (*accuracy, %*). RAVS achieves state-of-the-art results across all three benchmarks, outperforming the strongest baselines by 2.53, 8.85, and 6.51 percentage points respectively.

Table 3: Ablation results. w/o Bidirectional denotes unidirectional reasoning only, w/o Reflection denotes no reflection mechanism, and w/o Second-pass Retry denotes only one attempt per problem.

Methods	Total	L1	L2	L3	L4	L5	L6
w/o Bidirectional	92.80	98.68	97.12	92.79	92.00	75.44	40.43
w/o Reflection	93.83	99.21	97.76	93.24	93.33	78.95	46.81
w/o Second-pass Retry	90.57	97.62	96.49	91.44	83.33	70.18	38.30
RAVS	96.06	99.74	98.72	94.59	96.00	91.23	61.70

Table 3 report comparison results. It can be observed that: (1) Bidirectional reasoning contributes on hard problems: removing it costs 15.79% on L5 and 21.27% on L6, compared to only marginal drops 3.26% on L1. This confirms that bidirectional reasoning is crucial for solving difficult problems. (2) Reflection and memory contributes 12.28% on L5 and 14.89% on L6, while showing negligible impact on L1, confirming that the reflection mechanism is crucial for preventing dead-end search in long-chain reasoning. (3) Second-pass retry contributes 21.05% on L5 and 23.40% on L6, yet only 2.12% on L1, serving as an effective recovery mechanism specifically for hard problems. These results demonstrate that all three components are essential, particularly for high difficulty problems.

Statistical information such as tool calls during the interaction process is shown in Figure 4. Most failed problems hit the interaction limit, resulting in balanced resource consumption across difficulty levels. For solved problems, difficulty positively correlates with resource usage. From L1 to L6, interaction count, context length, and tool calls all increase, indicating that complex problems require longer reasoning chains, richer context, and more frequent tool assistance.

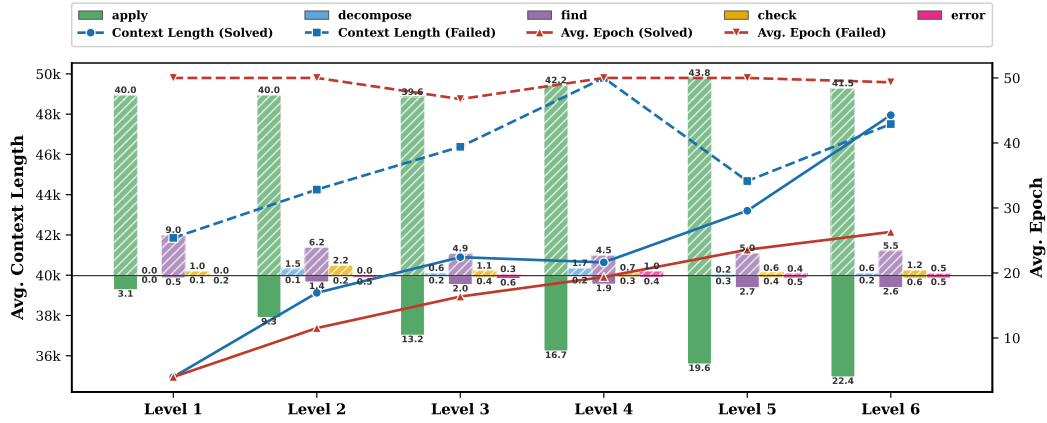


Figure 4: Statistical analysis of the interaction process. The bar chart represents the average number of tool calls, while the line charts represent the average context length and the average number of interactions. Dashed lines represent failed problems, while solid lines represent solved problems. For solved problems, difficulty positively correlates with resource usage.

6 Conclusion

We present RAVS, a unified neuro-symbolic framework that achieves 96.06% solving accuracy on FormalGeo7K by decoupling LLM planning from symbolic execution. A bidirectional reasoning engine with proven exponential efficiency advantages unifies forward and backward solving, while an agentic loop enables adaptive reflection. RAVS surpasses prior methods without any task-specific annotated data, demonstrating the power of principled neural-symbolic integration.

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